# Diffusion-Based Memory: Controlled Recall Across Semantic Scales

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## Abstract

Retrieval Augmented Generation (RAG) pipelines have become the dominant method to manage long-term memory in AI agents, addressing context window limitations through external knowledge base. While this is effective for factual recall, RAG treats memory as a static repository of retrievable facts, where either the memory is full retrieval or not at all. This is a coarse way to manage memory. We explore specifically how to model smooth and controlled recall of a memory item. This enables two main capabilities of the agents: 1) early exit in the generation process, where the agent can terminate the retrieval process once the sufficient information has been surfaced; and 2) selective recall, where details retrieved are contextually faithful and controlled. We study the first in the context of generation of general details of text, whereas the latter in the retrieval of privacy information. Our results show that while generating details in the smooth monotonic increase is possible, it is not always accurate. Whereas, controlled retrieval of details based on a context is not trivial, and requires a more in-depth understanding of the noise schedule.

## 1 Introduction

### 1.1 Problem and motivation

As AI agents are increasingly expected to support long-term interaction, personalization, and continuity across sessions, the question of how they should remember becomes more important. Most existing memory mechanisms are built around storage and retrieval, like Retrieval Augmented Pipelines (RAG), where information is preserved externally and fetched when needed. While this is effective for factual recall, it treats memory as a static repository of retrievable facts, where either the memory is full retrieval or not at all, with no mechanism of partial recall or compositional reconstruction. This is a coarse way to manage memory.

This motivates the central problem in our project: whether memory in language systems can be modeled so details can be recalled from abstracted texts, in a smooth and controlled manner Smooth here refers to monotonic increase in detail recall and controlled means selective recall based on some context. The core intuition is that not all parts of a memory are equally important at all times. Some information should remain available in a more durable form, while other information may reasonably weaken.

Therefore, we pose two problems:

- Progressive Recall: the problem of reconstructing a memory such that each successive recall step strictly refines the previous one, adding new detail without contradicting prior detail. Solving this problem can allow agents to exit early in the generative process, generating only the information needed. This can optimize token use and help manage context window limitations.

- Selective Recall: Building on the above problem, we also want to be able to control the type of attributes being recalled. For example, relevant Personally Identifiable Information (PIIs) based on context. Solving this problem will help build agents that are have hold contextual integrity in their responses (Nissenbaum, 2004).

### 1.2 Proposed Approach

We approach the problem of modeling smooth and controlled recall in AI agents in two different methods, which involve two different paradigms:

1. Progressive Recall: Through the method of generation on a given starting memory, the model is able to recall details, like names, numbers, and locations, in a monotonic manner. A continuous latent diffusion model is

trained to generate language in the reverse diffusion step, such that each reverse step corresponds to progressive recall of information.

2. Selective Recall: For this problem, we use a different paradigm, and we explore how the model itself can be used as a memory system, where the training data can be returned verbatim, or close approximations. And this is possible because when trained, the model parameters encoding training data which can be used for recall. And specifically, selective recall based on the contexts.

Figure 1 illustrates the full REMIND pipeline on a concrete example. The two phases, forward diffusion as graded forgetting, and reverse diffusion as cue-conditioned recall, operate symmetrically in the latent space but serve distinct purposes. During forward diffusion, the per-subspace noise schedule $\beta_k(t)$ progressively destroys episodic detail in $v_0$, producing a sequence of increasingly abstract representations that can be decoded into natural language at any noise level. During reverse diffusion, the gist cue $u$, a compressed semantic summary of the memory encoded by the subconscious head conditions each denoising step, shaping which details re-emerge and in what order. The result is a recall trajectory that mirrors the forward abstraction in reverse: coarse semantic content surfaces first, with finer episodic detail recovered only as denoising continues. This structure is what distinguishes REMIND from standard retrieval pipelines, which return a fixed memory snapshot independent of how much detail the current context actually requires.

### 1.2.1 Generative Progressive Recall

Our proposed approach models each memory item using two complementary latent views: a gist-oriented latent *u* for relatively stable semantic structure, and a detail-oriented latent $v_0$ for more specific episodic information. The pipeline has three stages. First, the input is encoded and compressed, and two heads produce the gist and detail latents. Second, forward diffusion is applied to the detail latent to create noisier states corresponding to increasing forgetting levels, and a denoiser is trained to reverse that corruption. Third, a semantic projection module $G_\psi$ maps the denoised representation, together with the noisy latent, timestep, and gist cue, into a decoder-compatible space from which

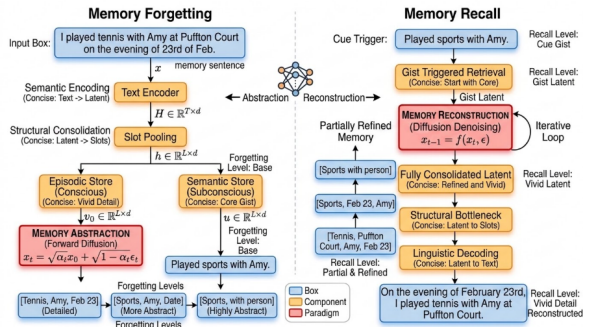



Figure 1: The REMIND architecture illustrated on a representative memory item. **Forward diffusion** (top) progressively abstracts an episodic memory across noise levels: a specific event — "Mr. Steve reacted to shellfish at an Italian restaurant on Saturday" — is compressed first to a gist-level paraphrase, then to a generic category label ("A customer is allergic."), simulating natural forgetting over time. **Reverse diffusion** (bottom) inverts this process conditioned on a semantic cue $u$ ("What was Mr. Steve allergic to?"), progressively recovering episodic specificity: the recalled output reintroduces the individual, the allergy type, and the event context in a coarse-to-fine order

text can be generated. This staged design was intended to separate representation learning, corruption and recovery, and final generation.

### 1.2.2 Diffusion Memory System for Selective Recall

The proposed diffusion memory system is grounded in a reframing of the above architecture, where rather than treating memory as an external store accessed by the model, we propose that the model itself is the memory system. Detail recall are not delegated to a separate retrieval pipeline but the model itself effectively stores and returns its training data, as the model's parameters end up encoding information about that data. So at inference time, those training examples or close approximations of it can be reproduced. This involves understanding model memorization in diffusion models, which is the mechanism that makes "the model as memory store" viable.

The training approach is similar to the above Generative Memory Recall approach of using latent diffusion for language generation, with some slight modifications. It doesn't have projection of memory in to different latents like semantic and episodic, and no projection module because we are not concerned with generation quality, instead the goal is to encode the training data to recall, in a memorized manner. So the architecture is

more simple, with just the encoder-decoder trained to project it to latent space (to make project text (discrete) to a continuous space), and continuous reverse diffusion is trained to remove noise step by step, which should be controlled.

## 2 Related Works

Our work draws on several lines of research spanning abstractive compression, diffusion-based language modeling, associative memory, retrieval-based memory systems, and memorization analysis. The original architecture was motivated by the idea of representing a memory item at multiple semantic scales, which makes abstractive compression a natural starting point. Toutanova et al. (2016) provide a dataset and evaluation framework for abstractive compression of sentences and short paragraphs, which made the MSR setting useful for our early experiments on semantic reduction. However, compression datasets do not themselves model progressive forgetting, intermediate memory states, or reverse-time recovery, so they serve better as an initial environment than as a complete memory benchmark.

On the modeling side, our original architecture also builds on encoder-decoder generation and controllable representation learning. Raffel et al. (2020) show that many NLP tasks can be formulated in a unified text-to-text framework, making encoder-decoder models a natural basis for conditional generation. In addition, work on disentangled and controllable text representations, such as Bao et al. (2019) and Hu et al. (2017), supports the idea that different aspects of language can be separated in latent space rather than forced into one undifferentiated representation. This perspective informed our decomposition into gist-oriented and detail-oriented latents.

The mathematical foundation of the project comes from diffusion modeling. Ho et al. (2020), Nichol and Dhariwal (2021), Song et al. (2020), and Song et al. (2021) together establish diffusion as a framework of gradual corruption and learned reverse denoising, which directly supports our treatment of forgetting as forward corruption and recall as reverse recovery. At the architectural level, Peebles and Xie (2023) show that diffusion can be implemented effectively with transformer backbones, which is especially relevant in our setting because the denoised objects are structured language representations. Recent work on diffusion for language further strengthens this connection: Li et al. (2022), Lovelace et al. (2023), and Shabalin et al. (2024) show that diffusion can operate meaningfully over continuous language representations and language-model encodings. These papers are particularly relevant to the current stage of our project, where the new architecture studies privacy-rich records through diffusion-style corruption and recovery in continuous text space.

The conceptual interpretation of diffusion as memory is informed by associative-memory work. Ramsauer et al. (2020) connect modern Hopfield networks with transformer attention, while Ambrogioni (2024) argue that diffusion models themselves can be viewed as associative memory systems. These works were important for our original framing because they justify viewing denoising trajectories not only as generation procedures, but also as movement through a recoverable memory landscape. Our work adopts this view but extends it toward graded abstraction, privacy-sensitive recall, and memorization analysis in text diffusion models.

Retrieval-based systems provide an important comparison point. Lewis et al. (2020) formulate retrieval-augmented generation as a hybrid of parametric and non-parametric memory, and Blattmann et al. (2022) show that diffusion generation can also be conditioned on retrieved neighbors. These works clarified that compactness or storage efficiency alone would not be a sufficient contribution against strong retrieval-based baselines. Similarly, downstream benchmarks such as Maharana et al. (2024) and Zhao et al. (2025) helped clarify what realistic memory evaluation could look like, while also highlighting the scale and complexity of broad conversational-memory setups. This helped motivate the shift toward a narrower privacy-oriented question.

The most directly relevant literature for the new architecture is the memorization and privacy-leakage literature. Carlini et al. (2021) show that autoregressive language models can leak verbatim training examples, including personally identifiable information, under black-box extraction attacks. The closest direct prior work for our current scope is Luo et al. (2026), which studies memorization in diffusion language models and argues that prefix-based extraction does not fully capture their native denoising process. They propose a generalized extraction framework for arbitrary masking patterns and stochastic denoising, making this paper espe-

cially important for our current evaluation design.

Finally, the move toward Privasis is grounded in privacy-sensitive data and evaluation work. Kim et al. (2026) introduce a large synthetic private dataset and privacy-oriented benchmark suite, giving us a corpus in which both full-record leakage and attribute-level leakage can be measured explicitly. Combined with multi-dimensional evaluation work such as Zhong et al. (2022), this supports a view of privacy-sensitive generation in which memorization must be evaluated not only through exact copying, but also through structured leakage and recoverability under different attack conditions. Taken together, these prior works shaped the project in two stages: an original architecture for graded semantic abstraction, and a newer privacy-oriented direction focused on memorization, extraction, and diffusion-native privacy risks.

## 3 Method

### 3.1 Progressive Recall

First, we look at the problem of progressive recall. The architecture consists of two main phases: P0 (latent representation) and P1 (diffusion-recall). P0 has five primary components: a transformer-based text encoder, a slot-pooling bottleneck, subconscious/conscious latent heads, a semantic projection module and a transformer-based text decoder. The overall training procedure is structured in a pipeline fashion. First, the encoder, slot pooling bottleneck, and latent heads are trained to produce the semantic latent $u$ and the detail latent $v_0$. Next, forward diffusion and denoising are defined over $v_0$ to model forgetting and latent recovery. Finally, the semantic projection module and decoder are used to map latent states at different timesteps back into readable text.
P1 has three primary components: a forward diffusion block, a denoiser block and a one-step latent estimation block. P1 also reuses the semantic projection module and text decoder from the P0 phase. We will go into the details of each component in the subsequent sections.

#### 3.1.1 Text Encoder and Slot Pooling

The text encoder's primary function is transform the memory item from a sentence representation into the embedding space. Mathematically, this implies that given an input memory item $x$, the encoder produces $H$, a sequence of hidden states, of dimension $T * D$, where $T$ is the number of tokens in $x$ and $D$ is the internal dimension of the encoder.
Because $T$ varies across memories, we compress $H$ into a fixed-size bottleneck via *slot pooling*:

$$h = \mathrm{CrossAttn}(Q,\ K{=}H,\ V{=}H), \quad h \in \mathbb{R}^{L\times d}, \tag{1}$$

where $Q \in \mathbb{R}^{L\times d}$ is a set of $L$ learnable query vectors, each specialised to capture a different facet of the sentence. This is preferable to mean-pooling because structural composition across the $L$ slots is explicitly maintained.

The encoder is trained with a contrastive objective (InfoNCE) that pulls the representation of $x$ close to a semantics-preserving perturbation $x^+$ (generated offline by Gemma) and pushes it away from representations of unrelated sentences in the same batch.

#### 3.1.2 Subconscious and Conscious Latent Heads

We split the output representation $h$ from slot pooling into a "gist" $u$ and "details" $v0$, through the use of two separate subconscious and conscious heads. The motivation behind this is to set aside the gist for cue-conditioned recall during the recall phase, while the details can be decoupled to move ahead through the diffusion process (for forgetting).
Mathematically, this would read as the representation $H$ is split into:

1. **Semantic Core ($u$):** Captures invariant semantic meaning of the sentence.
2. **Episodic Latent ($v_0$):** Captures episodic/detail through specific wording, entities, numbers.

Two linear projection heads are applied to $h$ to extract the two information tiers:

$$u = \mathbf{W}_u h + b_u, \quad u \in \mathbb{R}^k, \tag{2}$$

$$v_0 = \mathbf{W}_v h + b_v, \quad v_0 \in \mathbb{R}^{L\times d}. \tag{3}$$

$u$ is a compact, fixed-size vector that encodes the gist of the memory item. $v_0$ is a slot-structured representation of the details of the memory item trained to reconstruct the original sentence $x$:

$$\mathcal{L}_{\text{rec-x}} = -\log p_\theta(x \mid v_0). \tag{4}$$

The two heads are encouraged to specialise: $u$ retains stable semantics while $v_0$ retains recoverable

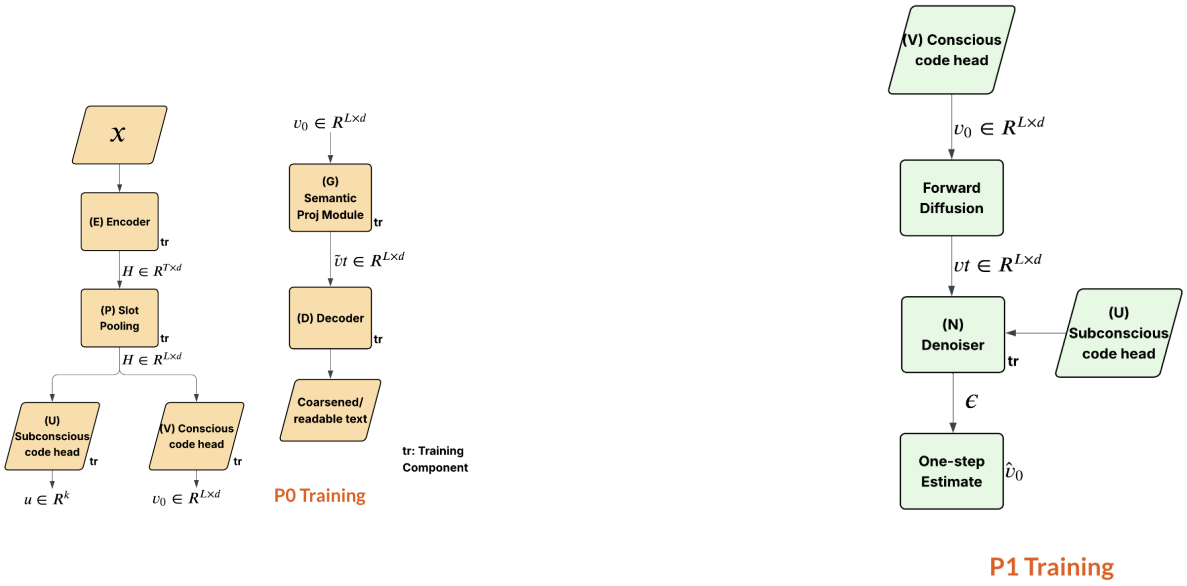



Figure 2: Stage-wise view of the architecture. P0 learns the latent memory representation, P1 models corruption and latent recovery over the detail latent.

specifics. The combined encoding loss is:

$$\mathcal{L}_{\text{enc}} = \mathcal{L}_{\text{NCE}} + \lambda_x \mathcal{L}_{\text{rec-x}} \tag{5}$$

with $\lambda_x as a balancing hyperparameter$.

To depict this with an example, consider the following memory item:

> *At 8pm last Saturday, John went to Taco Bell and had an allergic reaction to the shellfish I ate there*

The semantic core or "gist" of this sentence should read as:

> *John is allergic to shellfish*

and the episodic memory or "details" of this sentence should read as:

> *John had an allergic reaction to shellfish at Taco Bell at 8pm last Saturday.*

This separation enables the details to then go through forward diffusion (explained in subsequent sections), where it gets progressively noised (forgotten), while the gist stays stable to enable having an anchor for the cue-conditioned recall of the noised details.

#### 3.1.3 Semantic Projection Module

To generate readable text at any timestep, we introduce $G_\psi(v_t, u, t)$, a semantic projection module. This module integrates the noised detail latent $v_t$, the stable gist $u$, and the timestep $t$ to produce a projected latent $\hat{v}_t$ in the decoder input space.

Implemented along Adaptive Layer Normalisation (AdaLN):

$$\text{AdaLN}(v_t) = \gamma(u,t) \odot \text{LN}(v_t) + \beta(u,t), \tag{6}$$

where $\gamma, \beta$ are learned affine parameters produced by an MLP from $(u, t)$ preventing distributional mismatch between the noised representation and the decoder's input space

#### 3.1.4 Text Decoder

The output $\hat{v}_t$ from the semantic projection module is fed into a transformer decoder to produce a text representation $\hat{y}_t$, representing coarsened, but readable text at forgetting level $t$, which can then be compared to ground truth from the dataset.

$$\hat{y}_t = D(\hat{v}_t), \tag{7}$$

The semantic projection module $G_\psi$ is not supervised through a separate standalone objective. Instead, its learning signal comes through the decoder: given the timestep-conditioned latent input and the semantic cue $u$, $G_\psi$ produces a projected

representation in the decoder's input space, and the decoder is trained to generate the reference text associated with forgetting level $t$. Thus, the decoder loss also serves as the supervision for $G_\psi$, encouraging the projection module to map noised or partially recovered latent states into representations that are both semantically aligned and decodable. Concretely, if $x_t$ denotes the target text at forgetting level $t$, the decoder is trained with a cross-entropy objective of the form

$$\mathcal{L}_{\text{dec}} = -\log p_\theta(x_t \mid \hat{v}_t), \tag{8}$$

so that $G_\psi$ and the decoder are jointly optimized through the same timestep-specific generation target.

#### 3.1.5 Forward Diffusion

Following DDPM (Ho et al., 2020), we define a discrete-time Markov chain over $T$ steps that gradually corrupts the detail latent $v_0$:

$$q(v_t \mid v_{t-1}) = \mathcal{N}\Big(v_t;\ \sqrt{1-\beta_t}\, v_{t-1},\ \beta_t \mathbf{I}\Big), \tag{9}$$

where $\{\beta_t\}_{t=1}^T$ is the noise schedule. Crucially, this chain admits a tractable closed-form marginal at any timestep $t$ directly in terms of $v_0$:

$$\boxed{q(v_t \mid v_0) = \mathcal{N}\big(v_t;\ \sqrt{\bar{\alpha}_t}\, v_0,\ (1-\bar{\alpha}_t)\mathbf{I}\big)}, \tag{10}$$

or equivalently, using the reparameterisation:

$$v_t = \sqrt{\bar{\alpha}_t}\, v_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{11}$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. As $t \to T$, $\bar{\alpha}_t \to 0$ and $v_t \to \mathcal{N}(\mathbf{0}, \mathbf{I})$: the detail latent is fully destroyed.

**Forgetting as signal attenuation:** The signal-to-noise ratio (SNR) at timestep $t$ is:

$$\text{SNR}(t) = \frac{\bar{\alpha}_t}{1-\bar{\alpha}_t}. \tag{12}$$

Each timestep $t$ corresponds to a *forgetting level*: small $t$ weakens fine-grained lexical detail while leaving structure intact; large $t$ erases relational and event-specific content. The gist code $u$ is *not* noised, acting as a stable semantic anchor throughout.

**Noise schedule:** We adopt the cosine schedule from Improved DDPM (Nichol and Dhariwal, 2021):

$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, \qquad f(t) = \cos^2\left(\frac{t/T + s}{1+s} \cdot \frac{\pi}{2}\right), \tag{13}$$

with offset $s = 0.008$ to prevent $\bar{\alpha}_t$ from collapsing to zero too early. Compared with linear schedules, the cosine schedule exhibits a gentler SNR decay at small $t$, preserving gradual detail attenuation rather than abrupt destruction.

#### 3.1.6 Recall via Reverse Diffusion

The learned reverse process inverts the forward chain by progressively mapping a noised latent state to an earlier, less corrupted latent state:

$$p(v_{t-1} \mid v_t, u) = \mathcal{N}(v_{t-1};\ \mu(v_t, t, u),\ \Sigma(v_t, t, u)). \tag{14}$$

Following Improved DDPM (Nichol and Dhariwal, 2021), we parameterise the mean using the noise prediction $\hat{\epsilon} = \epsilon_0(v_t, t, u)$:

$$\mu(v_t, t, u) = \frac{1}{\sqrt{\alpha_t}}\left(v_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\hat{\epsilon}\right), \tag{15}$$

**One-step clean latent estimate.** At each reverse step we compute an immediate estimate of $v_0$:

$$\hat{v}_0 = \frac{v_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\epsilon}}{\sqrt{\bar{\alpha}_t}}, \tag{16}$$

which is used by the Semantic Projection Module to project back onto the encoder manifold and enable text decoding at any intermediate step.

**Multi-step latent estimate.** In addition to the one-step clean latent estimate, we also consider an iterative reverse process that progressively removes noise one timestep at a time. Instead of directly predicting $\hat{v}_0$ from $v_t$, we initialize the reverse trajectory with

$$\hat{v}_t = v_t, \tag{17}$$

and recursively estimate earlier latent states using the reverse-process mean:

$$\hat{v}_{s-1} = \frac{1}{\sqrt{\alpha_s}}\left(\hat{v}_s - \frac{\beta_s}{\sqrt{1-\bar{\alpha}_s}}\,\epsilon_0(\hat{v}_s, s, u)\right), \tag{18}$$

where $s = t, t-1, \ldots, 1$

This produces a sequence of progressively denoised latent states $v_t, \hat{v}_{t-1}, \hat{v}_{t-2}, \ldots, \hat{v}_0$, which can be passed through the Semantic Projection Module and decoder at intermediate timesteps to analyse progressive recall. In contrast to the one-step estimate, which directly predicts a clean latent from $v_t$, the multi-step estimate preserves the full reverse trajectory and is therefore better suited for studying whether recalled memories become gradually more detailed as denoising proceeds.

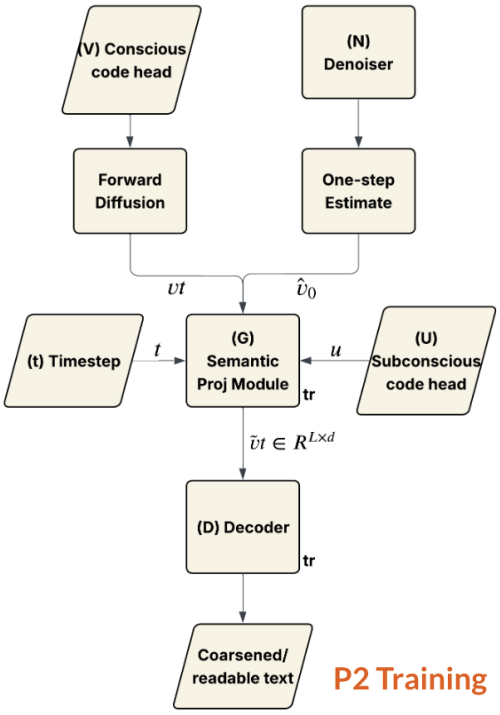



Figure 3: One-step latent estimation during recall. The denoiser predicts the noise in $v_t$ and directly estimates a clean latent in a single reverse step.

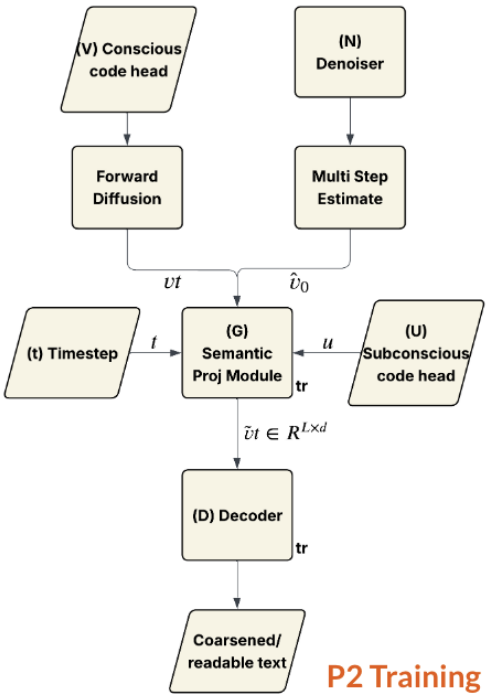



Figure 4: Multi-step latent estimation during recall. The denoiser is applied iteratively to move from $v_t$ to progressively less corrupted latent states, enabling step-by-step recovery.

### 3.2 Selective Recall

The method to solve selective recall problem builds on top of the work done in Progressive Recall. This method also involves training an BERT-based encoder-decoder architecture and the unconditional continuous reverse diffusion where the noise added is Gaussian. However, we shift to the paradigm of using the model itself as a memory system. In this case, the goal is memorization of the training data by model. While the ideal goal is to condition the recall of PII based on context such that the PII is relevant for that context only, we first simplified it to only recalling PII which could be in any order, not necessarily in contextual faithful way. In this method, we start the denoising step with a noised clean latent, and progressively remove noise to recover PIIs.

Therefore, these were the following main changes to the "Progressive Recalls" method:

1. The episodic and semantic latent head projections are removed since we are no longer using the gist to condition the recall, as we rely on memorization of training data.

2. The diffusion model is trained generate from noise, instead of from a degraded latents as in the case of progressive recall. This is to simply the architecture, and keep it close to the state of the art, like (Lovelace et al., 2023).

3. The inference is done on samples from the training data as the model is expected to have memorized and encoded the training data, which we are training to recall. This way the recovered PIIs should be exact or close-approximations of the PIIs in the training data.

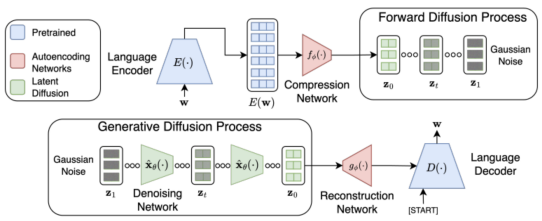



Figure 5: Architecture for Selective Recall

## 4 Experimental Setup

### 4.1 Datasets

**Abstractive Text Compression** We will use the MSR Abstractive Text Compression Dataset

(Toutanova et al., 2016) released by Microsoft, for progressive recall problem. The dataset is publicly available and contains original sentences/short texts paired with shorter abstractive compressions, with up to five reference compressions per input, along with human quality judgments such as meaning preservation and grammaticality. We will preprocess the raw files into our training format by loading the provided input-compression pairs, tokenizing them for the encoder/decoder, and generating $x^+$ examples during training as a semantic perturbation by removing 15% of characters from the text; these perturbations will be generated on the fly each epoch rather than cached. For splitting, we will use the official train/validation/test split provided with the dataset, and this off-the-shelf split satisfies our needs while keeping comparisons standardized

**Privasis-Zero** For the selective recall method, we use Privasis (Kim et al., 2026) as the primary data source because it provides privacy-rich natural language records together with structured attribute annotations, making it well suited for memorization in models. In each item, the main field of interest is the original record, which serves as the privacy-bearing text sequence, while fields such as attributes and grouped_attributes provide structured supervision about what sensitive information is present and how it is organized. This is especially useful for our project because the dataset is not limited to raw paragraphs alone; instead, it pairs each paragraph with machine-readable privacy information, allowing us to distinguish between direct identifiers, contextual sensitive details, and broader non-sensitive content. Since the currently accessible Privasis-Zero release exposes a single train split, we will construct a fixed 80/10/10 train/validation/test partition from that split for our experiments.

#### 4.1.1 Evaluation Methods

To measure the progress of both the proposed framework, we need to evaluate two main things, which is common for both:

1. The denoised memory at each step is monotonic.

2. The reconstructed memory after all the denoising steps is close to the input memory.

Since the two approaches use different methods, there is evaluation criteria specific to those methods.

For **progressive recall**, we further ensure that the reconstruction of the projected latent (v0) is as clean and as close as the input memory. For **selective recall**, we need to ensure that only the PII attributes are being recalled, and correct the given context.

The common metrics used to evaluate include BERT-score, UniEval and Detail Leakage Score.

**BERT-score** To evaluate sentence similarity, we will use BERT-score. BERT-score measures how similar a generated text is to a reference text by comparing their semantic meaning to evaluate at context level. The BERT-score is obtained by converting reference and candidate sentence into vector embedding using a pretrained transformer like BERT. Then the vector embedding of both sentences is compared using cosine similarity and F1 score. This is able to capture paraphrases and synonyms when calculating similarity.

**UniEval** In addition to the semantic similarity, we will use UniEval (Zhong et al., 2022) to further evaluate the coherence of the generated text. UniEval is a LLM based evaluation framework which has four evaluators. For our purpose we will use unieval-sum. While it is targeted for text summarization, we will use it for reconstructed text. The unieval-sum model across the following dimensions evaluates coherence, consistency, fluency and relevance which is aggregated.

**Monotonic Increase of Detail** To evaluate the effectiveness of the recalling aspect of the model, we need to measure the abstraction trajectory recalls detail monotonically. For this we use an entailment score. It is calculated using an natural language inference model on adjacent memory pairs. The idea is that a more detailed memory should support the previous more abstract one, but not the other way around. Therefore, for detail recall to occur, the backward direction should be entailment, meaning $s_{i+1}$ entails the content of $s_i$. And forward direction should not be entailment, meaning $s_i$ should not entail all the content of $s_{i+1}$. For each sample, the score is the count of entailment as percentage of the total degradation steps. And each samples scores is averaged, for a final score.

The evaluation set will consist of a test dataset from MSR (Toutanova et al., 2016). The test

dataset consists of 700 input memories, with forgetting trajectory (xt). Additionally, to evaluate the monotonic detail removal, we can further use the Privasis (Kim et al., 2026), which is privacy-rich text corpus dataset.

Therefore, we will conduct the following evaluation to measure progress:

1. To evaluate the reconstruction of the projected latent (v0), we will use BERT-score and UniEval between decoded projected latent (v0) and input memory.

2. To evaluate the monotonic forgetting of input memory, we will use the Detail Removal Score.

3. To evaluate the final reconstruction of memory after denoising, we will use BERT-score and UniEval between the reconstructed and the input memory.

### 4.2 Implementation details

#### 4.2.1 Off-the-Shelf Assets and Libraries

We used a pretrained Flan-T5 as our backbone for the encoder and decoder. The system is implemented in PyTorch with Hugging Face Transformers for backbone loading and tokenisation, Weights & Biases for experiment tracking, and SLURM for job scheduling on the Unity cluster.

The denoiser $\epsilon_\theta$ is a transformer with $N{=}6$ blocks, $n_{\text{heads}}{=}8$ attention heads, embedding dimension $d{=}512$, and FFN inner dimension $d_{\text{ff}}{=}2048$ ($4{\times}d$), with dropout $p{=}0.1$. It operates over $L{=}8$ memory slots and is conditioned on the gist code $u$ and diffusion timestep $t$ via AdaLN at every block. The forward diffusion uses a cosine noise schedule over $T$ steps.

#### 4.2.2 Computational Requirements

Primary training runs on an NVIDIA A100 (80 GB) via the `superpod-a100` partition, with 4 CPU cores and 80 GB memory-per-CPU allocated per job and a wall-clock budget of 20 hours. Storage is allocated as ∼50–100 GB covering pretrained weights, cached abstraction trajectories $\{x_t\}$, and rolling checkpoints.

## 5 Results

The central hypothesis of the original architecture was that if memory were modeled as a controllable, graded process, then the system would exhibit controlled forgetting and smooth recall over time. Our experiments provide mixed evidence. On the positive side, the model learns a meaningful detail-bearing latent and some architectural refinements improve local reconstruction quality. On the negative side, these improvements do not translate into reliably smooth forgetting or semantically well-ordered recall trajectories. The experiments therefore support only a weaker claim than originally hypothesized: the architecture learns useful latent structure, but does not yet produce the intended memory dynamics.

### 5.1 Progressive Recall Results

#### 5.1.1 One-step versus multi-step latent estimation

We first compared two reconstruction strategies during recall: one-step latent estimation and multi-step latent estimation. The one-step strategy directly predicts a clean latent from $v_t$, while the multi-step strategy iteratively denoises through intermediate latent states. The comparison is informative because it separates direct latent recovery from progressive reverse diffusion.

| Metric | One-step | Multi-step |
|---|---|---|
| BERTScore F1 | 0.8199 | 0.8251 |
| BERTScore Precision | 0.8547 | 0.8565 |
| BERTScore Recall | 0.7884 | 0.7966 |
| UniEval Consistency | 0.6432 | 0.6000 |
| Best Val Loss | 0.3165 | 0.4114 |
| Train Loss | 0.1304 | 0.4043 |
| Val Loss | 0.4294 | 0.4245 |

Table 1: Comparison of one-step and multi-step latent estimation. Multi-step improves BERTScore and final validation loss, while one-step achieves higher UniEval consistency and a lower best validation-loss snapshot.

Table 1 and Table 2 show that the comparison is mixed rather than uniformly favorable to one method. Multi-step achieves slightly better BERTScore F1, precision, recall, and final validation loss, while one-step achieves higher UniEval consistency and a lower best validation-loss snapshot. The qualitative examples nevertheless favor the multi-step strategy. For instance, in a particular example about comparing the evidentiary record and the financial picture, the multi-step run reconstructs a semantically meaningful fragment, whereas the one-step run collapses to a vague phrase such as “There is a view idea.” Likewise, for the “Climb up to the ruins” target in a different example, both strategies perform poorly, but

| Memory Level | One-step Output | Multi-step Output |
|---|---|---|
| **Target:** *"The evidentiary record and the financial picture together tell a more complicated story."* | | |
| Level 1 | "There is a view idea." | "The record and the financial picture tell a complicated story." |
| Level 50 | "There is a view." | "The evidence and financial picture together suggest a complex narrative." |
| Level 100 | "There is a view idea." | "The financial picture tells a story." |
| Level 250 | "There is an idea." | "There are complications." |
| Level 500 | "There is." | "There is a difference." |
| **Target:** *"Climb up to the ruins of the castle to admire the views."* | | |
| Level 1 | "Climb up to the castle." | "Climb up to the ruins to admire the views." |
| Level 50 | "Climb up to the ruins." | "Climb up to the ruins and admire the views." |
| Level 100 | "Go up to the castle." | "Go up to see the ruins." |
| Level 250 | "Go see the view." | "Visit a historic site." |
| Level 500 | "Go to the view." | "There is a place to visit." |

Table 2: Qualitative comparison of one-step and multi-step latent estimation across memory levels. The multi-step strategy retains more semantic content at intermediate levels, while the one-step strategy tends to collapse earlier to degenerate outputs.

the multi-step output remains somewhat less degenerate than the one-step output. Based on this combination of slightly stronger similarity scores and more semantically informative qualitative behavior, we selected multi-step latent estimation for subsequent experiments. At the same time, multi-step estimation is not a complete solution to the reconstruction problem; the gains are incremental rather than decisive. Nevertheless, the qualitative evidence in Table 2 suggests it is a more principled foundation for the controlled recall pipeline, and we carry it forward accordingly.

### 5.1.2 Quality of the detail latent

We evaluated whether the detail latent $v_0$ captures enough information to support later forgetting and recall. This is done by decoding $v_0$ directly and comparing the resulting text against the original sentence using BERTScore and UniEval-based quality measures. The aggregate result is encouraging: the distribution of BERTScore values for direct $v_0$ projection is concentrated in the high range, indicating that the detail latent often preserves substantial semantic information.

However, the examples in Table 3 show that high semantic overlap alone is not sufficient. In the first example, the decoded output remains close to the source and preserves factual content reasonably well. In the second, the BERTScore remains similarly high, yet factual consistency drops sharply, even though the prediction still appears superficially related to the original. This indicates that $v_0$

| |
|---|
| **Example 1** |
| **Ground truth:** This has led, says the Journal, to annual compensation rising 5–6 percent a year. |
| **Prediction:** This has led to, according the Journal, to say compensation rising 5% 6 percent a year. |
| **Metrics:** BERTScore F1 = 0.960, UniEval Factual Consistency = 0.939, UniEval Task Summarization = 0.736 |
| **Example 2** |
| **Ground truth:** This little museum celebrates the ingenuity and courage of those who sought to escape to the West and commemorates those who died trying to do so. |
| **Prediction:** This little museum museums the ingenuity and courage of those who sought to escape to the west, and commemorates those who do doing to do so. |
| **Metrics:** BERTScore F1 = 0.961, UniEval Factual Consistency = 0.135, UniEval Task Summarization = 0.682 |

Table 3: Representative examples of direct decoding from the detail latent $v_0$. High BERTScore does not always imply high factual consistency.

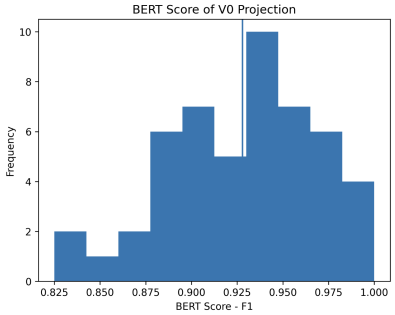



Figure 6: Distribution of BERTScore F1 for direct projection from the detail latent $v_0$. Most values lie in a relatively high-similarity range, suggesting that $v_0$ preserves substantial semantic information.

is learning useful episodic structure, but faithful decoding remains unstable. Thus, the latent representation itself is promising, but its downstream use in a controllable memory system remains unresolved.

### 5.1.3 Ablations

We also ran a set of design-decision experiments to test whether targeted architectural changes improved generation quality. These ablations were useful because they separated local component improvements from the larger question of whether the full system achieved smooth forgetting and recall.

| Experiment | Decision | Observation |
|---|---|---|
| Added $G_\psi$ in P0 and P2 | Adopted | Loss decreased and some validation samples became more similar to ground truth, although the improvement was modest. |
| Increased denoiser training to 1000 epochs | Adopted | Sentence quality in the P2 validation set improved relative to the earlier 100-epoch setting. |
| Partially unfroze $U$-head parameters in P2 | Rejected | BERTScore and UniEval remained almost unchanged, with a slight decrease in performance. |

Table 4: Design-decision experiments for the original architecture. Some component-level modifications improved local generation quality, but they did not resolve the larger issue of unstable forgetting trajectories.

Table 4 shows that some architectural modifications were directionally useful, especially the addition of $G_\psi$ and longer denoiser training. At the same time, these gains were local rather than decisive. They improved reconstruction quality in some cases, but did not validate the broader memory hypothesis. This distinction is important for interpreting the midpoint results. The architecture was not failing uniformly, but neither was it producing the intended timestep-wise behavior.

### 5.1.4 Do we achieve smooth forgetting and recall?

The central hypothesis of the project required more than local reconstruction quality. It required that degraded memories become progressively less detailed in a monotonic and semantically meaningful way, and that reverse diffusion would recover detail in an ordered manner. To test this, we evaluated monotonicity using two intrinsic metrics: a monotonic decrease score based on BERTScore against the most detailed sentence, and an entailment score computed on adjacent degraded pairs.

| Setup | Monotonic score | Entailment score |
|---|---|---|
| Separate training | 0.164 | 0.628 |
| Joint training | 0.325 | 0.587 |

Table 5: Final monotonic detail-loss results for the original architecture. Joint training improves the monotonic decrease score, while separate training performs slightly better on entailment.

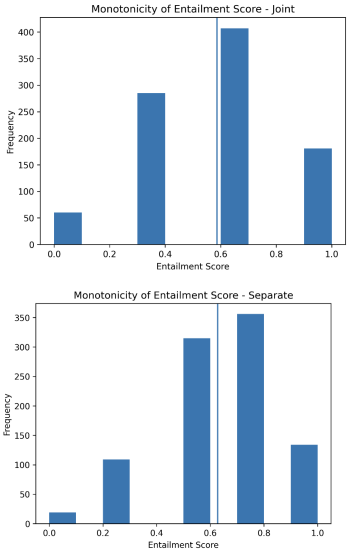



Figure 7: Representative monotonicity results for degraded memory trajectories.

The quantitative results in Table 5 do not support the strong version of the original hypothesis. Joint training improves the monotonic decrease score relative to separate training, but the values remain low in absolute terms, and the entailment scores do not indicate consistently strong monotonic behavior. The qualitative examples make the failure mode clearer. Table 6 shows a representative case in which the forgetting trajectory is not behaving as intended. In particular, the level-50 output is more detailed than the level-1 output, which violates the expected monotonic abstraction pattern. This failure is also reflected in the entailment result, which is false at the corresponding transition. More generally, Figure 7 shows that the model does not consistently produce a well-ordered coarse-to-fine trajectory across memory levels. Thus, the architecture can generate related text, but it does not yet realize the intended smooth forgetting behavior.

### 5.1.5 Summary of findings and motivation for the pivot

Taken together, the midpoint results support only a weaker claim than the one originally proposed.

| |
|---|
| **Ground truth:** Examples of condition information include, among others, averages of standardized condition rating codes; percentage of assets above, at, or below acceptable condition; or narrative information. |
| **Level 1:** "Examples include conditional conditional conditional codes" **Entailment:** – **BERTScore:** 0.881 |
| **Level 50:** "Examples include conditional conditional conditions, including standardized conditional codes, based on or below acceptable conditional conditions." **Entailment:** False **BERTScore:** 0.878 |
| **Level 100:** "Examples of information include conditional conditional attributes" **Entailment:** True **BERTScore:** 0.900 |
| **Level 250:** "Conditional terms are described as a measure of success" **Entailment:** True **BERTScore:** 0.866 |
| **Level 500:** "Efforts are given" **Entailment:** True **BERTScore:** 0.836 |

Table 6: Representative failure case for monotonic forgetting. The trajectory is not monotonic: the level-50 output is more detailed than level 1, and the first entailment check is also incorrect. This makes the example useful for illustrating both bad forgetting and bad entailment.

The model learns a useful detail-bearing latent, and some architectural refinements improve local reconstruction quality. Multi-step latent estimation also performs somewhat better than one-step estimation and was therefore selected for subsequent experiments. However, these gains do not translate into a convincing demonstration of controlled forgetting and smooth recall. The monotonicity results remain weak, the qualitative trajectories are often semantically unstable, and the current metric suite is not sufficient to fully validate the intended memory behavior.

For this reason, the results so far do not support the original hypothesis in its full form. Instead, they suggest that the architecture is better understood as a partially successful representation-learning system whose dynamics are not yet stable enough to justify the stronger memory claim. This motivated the pivot to the privacy-oriented direction based on Privasis. That setting is narrower and more testable: rather than asking whether the model supports smooth forgetting and recall in a broad semantic sense, it asks what information becomes memorized, leaked, or recoverable in a diffusion-based system trained on privacy-rich text.

## 5.2 Selective Recall Results

We next evaluate the system's ability to perform selective recall of privacy-sensitive information. Rather than smooth monotonic forgetting over time, this setting asks a more targeted question: can the reverse diffusion process recover specific personally identifiable information (PII) in a controlled and contextually faithful manner? The setup uses the Privasis dataset, which consists of 1.4M synthetic privacy-sensitive records spanning medical, legal, financial, calendar, and messaging domains. Forward diffusion is applied to compress and abstract these records, followed by reverse DDPM denoising to recover detail. We evaluate both one-step and multi-step estimation strategies, mirroring the ablation conducted for progressive recall. This setup is similar to (Lovelace et al., 2023).

### 5.2.1 One-step estimation for selective recall

Table 7 shows representative generation trajectories under one-step estimation across reverse diffusion steps, starting from full abstraction and progressing toward the original text. The target in the first example is a privacy-rich medical record containing a named individual, date of birth, blood type, and a specific fellowship application.

**Target:** *"Hi, this is Decorey Nguyen (DOB 2019-02-23, age 33, Vietnamese, A+, male). I just submitted my J-1 visa application for the Advanced Cardiovascular Imaging Fellowship on 2023-08."*

| Step | Generation |
|---|---|
| Early | "Project Date: March 9, 2022, assetsadobeDate: 2016-09-09. Llanger Family Statement That Stems the Assets of the Ontario Tourism Commission." |
| Mid | "Hi, I am Perorey Dong (DOB 2019-09-25, Age 17, male, age 79). I just submitted my VISA application for the Advanced Restoration Imaging Committee on 2023-08." |
| Later | "I just submitted my J-1 visa application for the Advanced Cardiovascular Imaging Fellowship on 2023-08." |
| Final | "Decorey, Phuley (DOB 2019-02-23, A+, male). I just submitted my J-1 Visa Application for the Cardiovascular Imaging Fellowship at the Crossroads Research Facility in 2023-08." |

Table 7: Representative one-step estimation trajectory for selective recall on a Privasis medical record. Early reverse diffusion steps are incoherent; mid-steps show fabricated PII; later steps recover the event structure with decreasing identity accuracy.

Several things are notable here. First, the final generation is clean, fluent natural language and correctly recovers the core event - a visa application for a cardiovascular imaging fellowship - along with a plausible date. Second, the trajectory is not entirely stable: the early reverse step produces an incoherent, largely hallucinated output, while the mid-step recovers the event structure but with fabricated PII (a different name, DOB, and institution). This fabrication behavior is not purely a failure mode; it indicates that the diffusion process learns to generate contextually plausible identities rather than simply copying or blanking them. The later step strips identity information almost entirely, recovering the event without the individual, which is itself a useful intermediate state for privacy-aware retrieval.

A second example (Table 8) illustrates a case where multi-step recovery is more erratic, but the

final output again converges on a semantically coherent reconstruction.

**Target:** *"2015-03-20 – After the Tuesday rehearsal at 19:30, I, Keagin Haugen (age 50, separated, lower-middle income), reviewed the upcoming Oslo segment for the mixed adult choir (38 members) and noted the lack of a dedicated travel budget for trips to St."*

| Step | Generation |
|---|---|
| Early | *[Garbled, incoherent output with fragmented tokens and non-linguistic characters.]* |
| Mid | "2015-03-20 – With a Tuesday morning rehearsal at 19:30, I, Keaton Benggen (age 50, lower-income), reviewed the upcoming Oslo Oslo conference for the first time (at least 50 years ago) and noted the lack of a dedicated travel room for the mixed force." |
| Later | "2015-03-20 – After the Tuesday rehearsal at 19:30, I, Keagin Haugen (age 50, separated, lower-middle income), reviewed the upcoming Oslo segment for the mixed adult choir (38 members) and noted the lack of a dedicated travel budget for trips to St." |

Table 8: Second one-step estimation example. The mid-step shows hallucinated PII (different name, altered income bracket, changed event type); the later step converges closely to the original record.

The mid-step here again shows hallucinated identity attributes - the name is altered to "Keaton Benggen," the income bracket is simplified, and the event type shifts from a choir rehearsal to a conference. Yet the date, time, and location cue (Oslo) are preserved. This pattern suggests the diffusion model encodes event-level structure more robustly than fine-grained identity attributes, which is consistent with the findings from progressive recall on the detail latent $v_0$.

### 5.2.2 PII abstraction behavior during reverse diffusion

A key observation across both examples is that PII is not handled uniformly during reverse diffusion. Table 9 illustrates two distinct behaviors that emerge at intermediate denoising steps: in some cases, PII is abstracted away entirely, leaving only the event structure; in others, PII is neither removed nor correctly recovered, but instead *fabricated* (replaced with a plausible but incorrect identity).

Both outcomes are informative for the privacy-oriented use case. Full abstraction is the desired behavior when the recall context does not authorize identity disclosure i.e the model surfaces the event without surfacing the person. Fabrication, while not correct, is preferable to direct leakage of the original PII: a downstream consumer would receive a plausible but unverifiable identity rather than a real one. The remaining challenge is *controlling* which behavior occurs, which requires a deeper understanding of how the noise schedule interacts with the encoding of identity attributes in the latent space.

| | Example 1 | Example 2 |
|---|---|---|
| **Original** | "Original UPN's potential breakout hit, Shasta McCrackhead, is tamed into the disastrously dull Shasta McNasty. Look where it is in the ratings now." | "Hi, this is Decorey Nguyen (DOB 2019-02-23, age 33, Vietnamese, A+, male). I just submitted my J-1 visa application for the Advanced Cardiovascular Imaging Fellowship on 2023-08." |
| **Abstracted target** | "A potential hit show was renamed and became less successful." | "I just submitted my visa application." |
| **Abstracted recovered** | "Progress potential show is designed to look dull significant." | "Hi, I am Perorey Dong (DOB 2019-09-25, Age 17, male, age 79). I just submitted my VISA application for the Advanced Restoration Imaging Committee on 2023-08." |
| **PII behavior** | PII fully abstracted; event gist preserved. | PII fabricated; name, DOB, and institution are plausible but incorrect. |

Table 9: Two distinct PII behaviors during reverse diffusion. In Example 1, identity information is abstracted away and the event gist is preserved. In Example 2, PII is neither removed nor correctly recovered, but replaced with fabricated attributes that preserve the surface form of the original.

### 5.2.3 Multi-step estimation for selective recall

Table 10 shows the multi-step estimation trajectory for a Privasis record describing a meetup event with location-specific PII. This example is representative of cases where multi-step denoising produces a more orderly coarse-to-fine recovery than one-step estimation.

**Target:** *"Meetup.com group chat archive for 'Isandlwana Heritage Reenactors'. Date: 2024-04-12. Location: Isandlwana Battlefield, R68, Nquthu, 3139, KwaZulu-Natal, South Africa."*

| Step | Generation |
|---|---|
| Earliest | *[Incoherent output: fragmented speaker-attribution tokens and non-linguistic repetition.]* |
| Early | "A group gathered to honor history through a reenactment." |
| Mid | "A group reenacted the Battle of Isandlwana at the historic battlefield." |
| Later | "A group of students from South Africa. Meetings were held at the Botswana Historical Museum." |
| Final | "Meetup.com group chat archive for 'Isandlwana Heritage Reenactors'. Date: 2024-04-12. Location: Isandlwana Battlefield, R68, Nquthu, KwaZulu-Natal, South Africa." |

Table 10: Multi-step estimation trajectory for a Privasis event record. The recovery moves from incoherence, through an accurate event-level gist, to a near-verbatim reconstruction of location-specific PII in the final step.

This trajectory is encouraging in several respects. The early step already produces a fluent, abstractive summary of the event - the reenactment context is correctly identified without any location-specific detail. The mid step refines this to name the specific historical battle, and the final step recovers the structured record including the date, road identifier, and postal code, with only minor discrepancies. The intermediate step that localizes to "Botswana Historical Museum" rather than the correct KwaZulu-Natal location illustrates that ge-

ographic specificity is one of the last attributes to stabilize, consistent with the idea that fine-grained location PII is encoded at high noise sensitivity in the latent space.

Taken together, the selective recall results suggest that the diffusion framework is a promising substrate for privacy-aware memory retrieval. The model demonstrates a meaningful separation between event-level structure and identity-level detail, and the multi-step trajectory produces a cleaner coarse-to-fine recovery than one-step estimation in most cases. The remaining open problem of controlling *which* PII attributes resurface and at which denoising steps is a tractable direction for future work.

## 6 Conclusion and Future Work

This work presented REMIND, a diffusion-based memory architecture for AI agents that models memory as a controllable, graded process rather than a binary retrieve-or-forget decision. The central idea that forward diffusion can simulate progressive forgetting and reverse diffusion can simulate cue-conditioned recall is well-motivated by both cognitive analogies and the practical limitations of existing retrieval-centric pipelines such as RAG. The experiments provide a nuanced picture of where this idea succeeds and where it falls short.

On the positive side, the architecture learns a meaningful detail-bearing latent representation. Direct decoding from $v_0$ produces outputs with consistently high semantic similarity to the source, indicating that the encoding stage captures substantial episodic content. Multi-step latent estimation outperforms one-step estimation qualitatively, producing more semantically informative outputs at intermediate denoising steps rather than collapsing to degenerate phrases. Several targeted architectural refinements of adding $G_\psi$ and extending denoiser training produced measurable improvements in local reconstruction quality. In the selective recall setting, the model demonstrates a meaningful separation between event-level structure and identity-level detail, and the reverse diffusion trajectory moves in a recognizable coarse-to-fine direction, recovering gist before specifics in many cases.

At the same time, the results do not yet validate the stronger version of the original hypothesis. Monotonicity scores for progressive forgetting remain low in absolute terms, and qualitative trajectories are often semantically unstable across noise levels. The entailment-based analysis reveals frequent violations of the expected coarse-to-fine ordering, indicating that the current noise schedule and training objective do not reliably impose a monotonic abstraction structure on the latent space. For selective recall, the behavior of PII during intermediate denoising steps, whether abstracted, fabricated, or leaked is not yet controllable, and determining whether memorization was successfully achieved in the Privasis setting remained inconclusive within the scope of this project.

These limitations point toward concrete directions for future work. On the progressive recall side, a more expressive monotonicity objective: one that operates directly in the latent space rather than on decoded outputs, may provide stronger inductive pressure toward ordered forgetting trajectories. On the selective recall side, targeted supervision at specific noise levels, rather than end-to-end reconstruction loss alone, is likely necessary to control which PII attributes resurface and when. The fabrication behavior observed at intermediate denoising steps is itself a potentially useful property: a model that replaces real PII with plausible but unverifiable surrogates offers a form of soft privacy protection that is strictly preferable to direct leakage, and warrants further study as a principled privacy mechanism.

More broadly, REMIND establishes that diffusion-based memory is a viable research direction for AI agent systems. The architecture demonstrates that graded, multi-scale memory representations can be learned from text, that reverse diffusion can recover meaningful semantic content from noisy latents, and that the framework naturally extends to privacy-sensitive settings where contextual control over recall is essential. Realizing the full potential of this approach - smooth, monotonic, cue-conditioned memory across semantic scales - remains an open problem, but the foundations laid here suggest it is an achievable one.

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